CHAPTER I

INTRODUCTION

This chapter introduces Land Grand, a proposed educational game for Grade 5 learners at Baluan Elementary School. It explains the project context, purpose, objectives, scope, and limitations.

Project Context

This section explains the background and learning concerns that guide the development of Land Grand. It begins with educational technology worldwide, then discusses home learning in the Philippines and the needs reported at Baluan Elementary School.

Educational technology should respond to the needs of learners and teachers. UNESCO's 2023 Global Education Monitoring Report recommends choosing technology that fits local conditions and supports fair access to learning. It also stresses the need for evidence and resources that can be maintained over time (UNESCO, 2023). These points guide the proposed use of a learning game at Baluan Elementary School.

The value of a digital resource depends on how it supports the lesson. In Land Grand, short explanations would be linked to activities involving landforms, rocks, and minerals. Learners would explore a game environment, collect items, and read descriptions related to the lesson. The project would examine this approach as an additional learning resource rather than assume that playing a game guarantees better understanding.

In the Philippines, printed and digital modules were among the learning options used during the COVID-19 period. The Department of Education's 2020 guidance allowed supporting materials, including interactive electronic resources, alongside distance-learning activities. It also described the role of adults who assist learners at home (Department of Education, 2020, Appendix A and Appendix B). This experience provides a national background for developing resources that support learning beyond the classroom.

Disasters can also affect the arrangements needed for learning at home. DOST-PHIVOLCS recorded a magnitude 7.8 earthquake offshore Sarangani on June 8, 2026 (DOST-PHIVOLCS, 2026). As of October 2026, Baluan Elementary School holds face-to-face classes. Modular learning is considered an option if another disaster interrupts classroom instruction. Land Grand is therefore intended to support home study during regular schooling and possible disruptions.

An initial survey at Baluan Elementary School on July 7, 2026 gathered responses from 20 parents and 20 teachers. Parent responses covered Grades 4–6, while teacher responses covered Kindergarten and Grades 3–6. The parent group was available during the scheduled parents' orientation meeting. Respondent selection was reported as randomized.

Parent responses showed difficulty explaining lessons in printed modules. In the parent group, 85% reported this concern. This finding supports considering additional explanations for home study. Land Grand would focus those explanations on the selected Science topics.

Teacher responses reflected mixed views about modular learning. Responses included doubts about how well it helped learners understand lessons. Other responses were positive or uncertain. The project therefore proposes additional activities alongside printed materials and teacher guidance.

Survey responses identified landforms, rocks, and minerals among the Science topics considered difficult. These reports provide a basis for selecting content for the game. The final lessons and selected items would still need content and curriculum validation.

The project also considers the use of examples during Science lessons. When physical rock or mineral samples are limited, digital illustrations can provide an additional way to present the material. Land Grand would combine these illustrations with names and descriptions of collected items. The digital examples would serve as supplementary learning material for the selected Science topics.

Most parents and teachers believed that a mobile game could support printed modules. However, some parents were unsure or unwilling to allow its use. Reported access to a smartphone, tablet, or computer also does not confirm access to a compatible Android device. These differences support offering the game as an additional resource while retaining printed materials and teacher guidance.

The survey describes the views of parents and teachers from several grades. It does not directly measure pupil achievement or the effectiveness of Land Grand. The Grade 5 target of the game is separate from the survey population. The findings provide local background for the proposed learning resource.

The project focuses on the following reported problems and potential material limitation:

1. Parents reported difficulty explaining printed module lessons to their children.
2. Teachers questioned whether modular learning helps learners understand lessons.
3. Survey responses identified landforms, rocks, and minerals as difficult Science topics.
4. Physical rock and mineral samples may be limited for classroom demonstrations.

These concerns support developing Land Grand while regular classes continue, before another disruption increases reliance on home study. The game would bring short lessons, item descriptions, and review activities into an offline resource. Learners could return to these activities across separate play sessions. This approach is intended to support home learning alongside the materials and guidance already provided by the school.

Purpose and Description of the Project

This section explains the purpose of Land Grand and describes its main learning activities. It shows how the proposed game would support Grade 5 learners alongside printed learning materials.

The game aims to support knowledge of landforms, rocks, and minerals through guided exploration. It would contain four episodes covering Mountains, Hills, Plains, and Plateaus. Before each level, learners would read a short lesson about the landform and selected rocks and minerals.

During play, learners would move and jump with an explorer character in a two-dimensional environment. They would collect rocks and minerals and read each item's

name and description. Obstacles and other challenges would form part of the exploration activities.

After a level, learners would see the collected items, a short review, and up to three gameplay stars. They could replay completed content or continue to the next available level. The game would save progress on the device for later sessions.

The project combines short explanations, interactive practice, and digital examples in one offline application. Digital examples would supplement physical specimens as additional visual material. The game is intended to support printed materials and teacher guidance during home study.

Objectives of the Project

This section presents the general and specific objectives of Land Grand. The specific objectives follow the same order as the project concerns and describe features that can be built and checked.

The general objective is to develop Land Grand, an offline Android exploration game for Grade 5 learners at Baluan Elementary School. It aims to support their knowledge of landforms, rocks, and minerals. Project completion is planned for November–December 2026.

The specific objectives are to complete the following within the planned period:

1. Provide short Science lessons before each level to support explanations during home study.
2. Implement interactive exploration and end-of-level reviews to supplement printed Science modules.

3. Develop four episodes covering Mountains, Hills, Plains, and Plateaus, including selected rock and mineral topics.
4. Present digital illustrations of selected rocks and minerals with names and short descriptions.

Scope and Limitations of the Project

This section describes the coverage and boundaries of the Land Grand project. It explains what the game includes and the limits of its use and the survey findings.

Scope

This section outlines the learners, content, game activities, and technical requirements included in Land Grand. Project completion is planned for November–December 2026.

The proposed game is for Grade 5 learners at Baluan Elementary School. Development would use Unity and C#, with Piskel and Clip Studio Paint for graphical assets. The application would include the following features:

- Four episodes covering Mountains, Hills, Plains, and Plateaus, with selected related rocks and minerals.
- Easy, Normal, and Hard modes with 3, 5, and 7 levels per episode, respectively. The proposed total is 60 levels across the four episodes.
- Short lessons before each level, digital illustrations with item names and descriptions during collection, and a review after completion.
- A two-dimensional game environment with large touch buttons, background music, and sound effects.

- Movement, jumping, and item collection, with challenges such as obstacles, traps, pitfalls, and time limits.
- Progression through difficulty modes and episodes, with replay, continuation to available levels, episode summaries, and a final completion screen.
- Up to three gameplay stars after a level. Detailed star and completion rules would be specified during development.
- A Main Menu with Play, Settings, Tutorial, and Exit, and a Landform Map for selecting unlocked episodes.
- Automatic saving and loading of the player's name, unlocked episodes, completed levels, earned stars, and collected items on the device.
- Offline use on Android phones and tablets running Android 10 and above. The minimum targets are 2 GB RAM and 300 MB free storage, subject to device testing.

Limitations

This section explains the limits of the game and how the survey findings should be understood. It identifies the boundaries of the project's learners, content, and offline use.

- The intended users are Grade 5 learners at Baluan Elementary School. The survey includes other grades and cannot be treated as evidence about all Grade 5 learners.

- Learning content is limited to the four selected landforms and selected rocks and minerals. Detailed content and curriculum alignment still require validation.
- The game excludes multiplayer, online accounts, online features, cloud storage, websites, and an administrator panel.
- Saved progress stays on the device being used. The project does not include central teacher monitoring or shared learner records.
- The learning activities do not include separate quiz screens, GPS functions, or real-world scanning.
- The game supports printed materials and teacher guidance. Stars, collected items, and completed levels are gameplay records rather than measures of learning gains.
- Digital illustrations are supplementary visual materials. They do not provide physical specimens or hands-on examination of real rocks and minerals.
- The initial survey describes views and expectations about learning support. Any claim that Land Grand improves knowledge requires a defined evaluation method and actual results.